Image Geolocation Challenge - Where in Europe Was This Photo Taken?

Praveen N Hegde
praveen.narayan.hegde@utn.de

ABSTRACT

I studied street-level image geolocation across 12 European countries under a network with a constraint of max 5 M parameter. After experimenting with various methods like spanning direct co-ordinate regression, hierarchical geo-cell classification I found that hybrid retrieval models performs extremely well. The `convnextv2_femto` model trunk with GeM pooling and 192-dimensional geographic descriptors using a distance-based soft-target KL loss. And at inference, I added methods like image whitening, hubness correction, Chamfer re-ranking, local geographic decoding and also it forms a gallery from labelled training set. And all these methods does not have any learnable parameters in them, so the total parameters count sums up to 4.92 M. So this whole network achieved 34.16km median haversine on 2,352 held-out images(mean 359.5km, 47.4% <25 km, 65.9% <200 km, 83.5% <750 km). This document reports all of the above contents, with proper numbers which prove the stated methods.

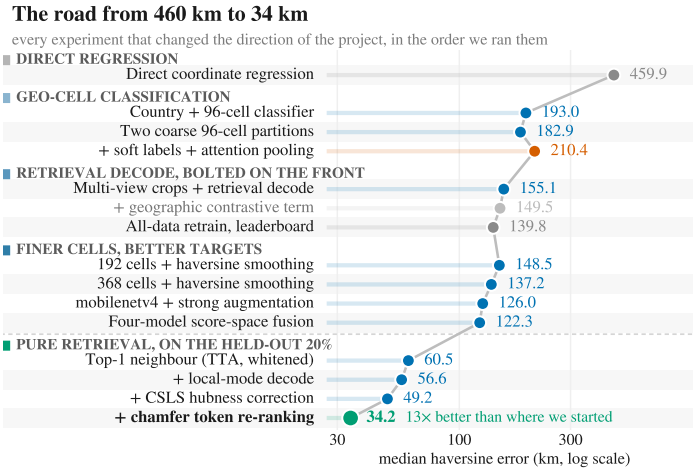


Figure 1: Major experiments across the 5 eras (shown as grey path by order of execution). Blue, vermillion and grey shows better, worse, and equivalent results compared to the prior one, on a per-era basis; green indicates the best one; The four eras above dashed pivot were on the internal protocol and cannot be directly comparable to held out result of submitted recipe on the lower one (§8)

1 Introduction

Image geolocation is confusing and difficult because similar looking images may come from different geographic location or different country and vise-versa. Within the constraints I had, achieving co-ordinate prediction and coarse regional classification is challenging. The following sections show why a single network was insufficient and how this led to a gallery-based retrieval system using embeddings from the 11,758 training images to estimate locations from nearest neighbours.

2 Experimental Progression and Key Findings

Figure 1 is the whole record, era by era; four conclusions from it shaped everything after.

- **Direct regression suffers structurally**, it is the most informative negative result generated. It turns out that a coordinate predicted by a small backbone model MobileVIT [1] is the conditional mean: the two most plausible rural locations for the image are eastern Poland or western France, thus placing the model's image optimally in central Germany! The issue was not overfit, but optimizing the wrong target. Simply classified the geo-cells [2, 3], which alone gained 267 km.
- **Finer cells and better targets kept paying, but only in tens of kms,** Two coarse partitions [4] beat one, since independent partitions cut finer than either one combined. So, 368 cells with haversine smoothed targets [5] beat 192, which both beat 96. A slimmer trunk (freeing parameterization for larger heads), over the fatter with small heads-again, better. All together-merging four of the previous partitions gives 122 km, in terms of the best of the era but again with 4x inference cost.
- **The gain came from the decoder, not the network.** The biggest single jump that period was an implementation for a multi-view crop decoding, in a retrieval mode, plugged into the cell classifier, but a geographic contrastive term on the same model proved to be of no significant help by itself. The system was already equipped with a retrieval system, preceded by the classifier.
- **Then it plateaued**: nine variations, no smaller than 122 km, so I stopped adding more and questioned what the floor was.

3 Diagnostic Analysis and Architectural Pivot

Each is minutes of arithmetic over cached predictions, no training. I should have run them months earlier.

Is the cell grid the limit? The perfect cells are classified, and predict cell center the errors are 85 km in cell 96, 53 km in cell 192 and 33 km in cell 368. As the model was about 150 km error with top 1 cell accuracy at 0.43 with cell resolution is still the limiting factor, not lack of "geographic info".

Is the answer even among the candidates? I evaluated how good the best candidate among the top-k retrieved neighbors was. Top-1 gave 109.7 km, while the best candidate among the top-32 gave only **8.63 km**; the closest labelled image anywhere in the gallery was 3.29 km away. This showed that the representation was already retrieving geographically close candidates, but the ranking was still weak. Therefore, the remaining effort focused on ranking and decoding rather than changing the network. The same analysis is shown in Figure 3 for the final model.

This analysis also exposed an earlier model-selection mistake. When I compared 11 sub-5M backbones without task-specific training, `convnextv2_femto` ranked first, with a 45 km advantage, while EfficientNet-B0 [6], which had been used throughout the earlier stages, ranked only fourth. It had been selected for classification and was never re-evaluated for retrieval.

4 Unsuccessful Approaches and Failure Analysis

These several experiments failed to provide useful results, instead demonstrating the constraints of the current setup and influenced the design of the final system.

- **Attention pooling:** inspired from [7], +38.6 km, where country accuracy also dipped and it was the only condition where this occurred. The reference used 8 transformer layers and 4.72m images; but I used one transformer layer and ≈9.4k images, so randomly initialized attention could not win at this size.
- **Haversine label smoothing** [5]: at $\tau = 50$, then $\tau = 100$ thrown away twice more as a broken mechanism - it was a tuning problem; tau needs to relate cell spacing (≈ NN-spacing/3), and at $\tau = 65/45$, it produced two highest gains of that period with one change. Cost approximately one layer of improvement.
- **Fixing France by supervision:** Country head plateaued at accuracy 0.78 (France 0.585), with nothing found for loss reweighting, MLP country head and gated decoding. This suggests the model is *confidently* incorrect, the real country wasn't even within the top-2.
- **Higher training resolution (512/576):** 30–60% more compute per epoch with no improvement over five runs, although it flip signs by objective — it *does* improve a starving backbone.
- **Horizontal flip augmentation:** turned off on purpose: mirroring eliminates driving side, sign text, and lane orientation; amongst other sparse cues distinguishing European countries, although the hypothesis was not studied.
- **Bundled changes:** Two experiments varied 3-4 parameters simultaneously. Both failed unattributably and produced misleading "it doesn't

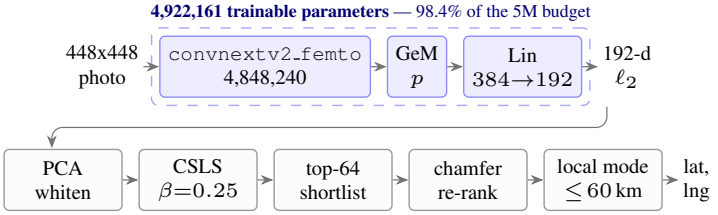


Figure 2: The whole system. The encoder (top) is everything that holds a weight; the decode chain (bottom) is arithmetic over stored vectors and adds no parameters. Whitening and hubness statistics are fitted on the gallery alone, so no query influences another.

work" verdicts that clean reruns later overturned. An EMA bug (decay 0.999, scored on the EMA weights) killed two more.

5 Final System Architecture and Methodology

Both diagnostics return basically the same result; the network isn't able to store a map of Europe (with less than 5M params) but is able to distinguish between two streets. That's the reason why I stopped to ask it *where was the photo taken* but I started asking *which photo this one resembles?* [8]

- That gets the map out of the weights and into the data Figure 2. Once all the labeled pictures are passed through the encoder, they are stored as 192-d unit vectors, one per picture, with their associated known locations, and are called the **gallery**. A new photograph is embedded the same way, and we read the answer off the nearest gallery vectors' coordinates. These stored vectors are data, not parameters: they hold no weights, receive no gradients, and count toward the 5M budget no more than the training images themselves do.
- **The encoder,** is the only part of the system that holds a weight – `convnextv2_femto` [9] with GeM pooling [10] and a linear projection, which sum up to **4,922,161 trainable parameters** in total. The backbone starts from ImageNet weights (`fcmae_ft_in1k`) and every parameter is then fine-tuned on the provided images; no other data of any kind is used.
- **While training:** For each batch, I convert the true geographic distances between images into a target distribution — closer images receive higher weights — and train the embedding similarity distribution to match it:

$$Q = \mathrm{softmax}_j(-d_{ij}/\tau), P = \mathrm{softmax}_j(e_i \cdot e_j/0.07), loss = \mathrm{KL}(Q\|P)$$

with self-similarity masked. P follows the standard InfoNCE form [11], while Q uses geographic distance. Each image acts as a positive according to how geographically close it is, so the model learns relative distance between neighbours instead of using a hard positive/negative split at a fixed radius [12].

- $\tau = 15$ km is by far the most critical figure for the whole project. There was a previously designed pipeline which worked at 250 km, where target would not be able to distinguish between 1 kmand 40 km, hence the embedding learns *region* and never *place* identity. Just this one value distinguishes between the 126 kmplateau and everything else.
- **Used geographically structured batch sampling:** Since only 24.6% of images have a partner within 500 m, random batches often contain no positive pairs and the loss degenerates. I therefore use 12 anchors × 4 images sampled within 25 km, which also provides hard same-region negatives. Training uses RandAugment [13] without horizontal flips (§4), AdamW [14] with OneCycle [15] for 40 epochs.

6 Inference and Coordinate Decoding

We employ four parameter-free stages to convert the nearest neighbor prediction into the coordinates as outlined in Figure 2. With the stages' individual gains are given in Figure 1 & Figure 3.

- **Multi-scale TTA,** (448/512/576, averaged and re-normalised) gives GeM pooling a finer feature grid at test time. **PCA-whitening** [16], fitted only on the gallery, removes dominant directions related to scene statistics, vegetation, and lighting that can suppress location-specific cues, giving my largest decoding gain of about 41 kmmore than every stage below

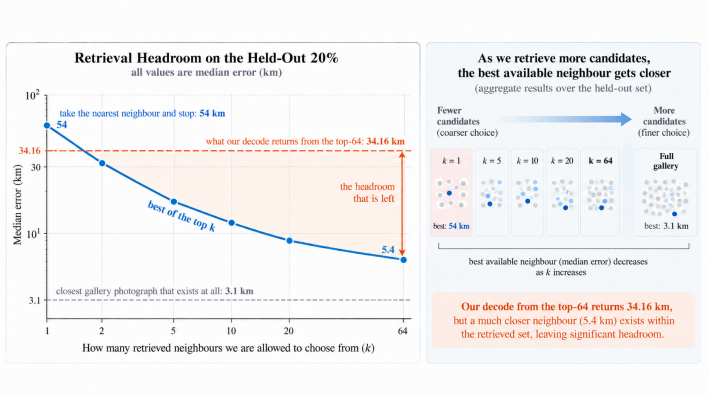


Figure 3: The diagnostic that redirected the project, re-run on the held-out 20%: the shaded band is headroom my decode does not take. The 34.16 km is that run's own full decode.

it combined. Restricting the spherical mean to a single **local mode** improves the result by another 3.9 km. **CSLS** [17] suppresses *hub* gallery images [18] and improves the result by a further 7.4 km by subtracting $0.25\times$ each gallery point's mean similarity to its 10 nearest gallery neighbours from its retrieval similarity score.

- **Chamfer re-ranking,** gives the largest gain within the ladder, -15.0 km. A global descriptor pools the full feature map, so visually different regions can still produce similar embeddings. I therefore compare spatial *token sets* over the top-64 shortlist using $\mathrm{score} = \mathrm{sim} + 3.0 \cdot \mathrm{cham}$, which checks whether local regions of two images match. This helps distinguish "another road in the same country" from "the same street". The method follows Patch-NetVLAD/DELG-style re-ranking [19, 20], and reuses the same projection for the tokens, so it adds no extra weights.
- **Neither half carries the result alone:** The decode chain without training in encoder yields 124 km; The trained encoder without decoding chain gives 109 km.
- **Also tried and dropped:** α-query expansion [10, 21] hurt performance by $+4$ km; k-reciprocal re-ranking [22], transductive whitening, and label propagation were neutral, while confidence-gated snapping gave 0.00 km gain. These methods only redistribute already retrieved neighbours, so they cannot help when the correct location is missing; CSLS is different because it acts before shortlist formation.

7 Remaining limitations

Two failures hide inside one median. If I look at what remains for a held-out of 20%, Iceland (6.9 km) Norway (7.2) Finland (11.0) Turkey (16.5) and Belarus (21.7) are solved while Sweden (33.2) Spain (33.7) and UK (71.0) are intermediate and Poland (135.2) France (402.9) Germany (448.7) and Italy (459.6) hold nearly all the remaining error.

It divides the two regimes. Re-ranking is not universally a good idea: Spain is reduced 60 → 34, Poland 228 → 135 (they had right street and so just mistook rank), but it makes things worse in France (369 → 403), Germany (413 → 449) and Italy (381 → 460). If there is no actual neighbours worth boosting, matching token sets can boost the wrong one. A frame from a French village looks like a hundred others, while one from an Icelandic valley is nearly unique. Those countries need coverage, not a better encoder.

8 Evaluation Protocol

The submitted model trains on all 11,758 labelled images, so it has no held-out set. I therefore validate the *recipe* using identical code on a country-stratified 80/20 split i.e. 9,406 images for training/gallery and 2,352 unseen images for evaluation, with all decode hyperparameters frozen and no search in `eval.py`. This gives **34.16 km** median error (359.5 mean; 47.4% <25 km, 65.9% <200 km, 83.5% <750 km); the final era of Figure 1 shows this evaluation stage by stage.

Three things I got wrong, and corrected.

(i) My early splits grouped by 500 m location cluster, removed nearby matches important for visual place recognition: 21.7% of images had

a co-located gallery twin under the official split, but 0.0% under ours. Therefore, the earlier eras in Fig. 1 show relative progress and are not directly comparable with the final result.

(ii) Tuning and reporting on the same data, cost 1.5 km where I measured it, at the earlier 59 km stage: 57.75 while tuning and scoring on the same predictions, 59.22 when the best config is picked on data held out from the tuning. I quote this one, and make all of the decode-related hyper-parameter fixed when taking on the 20% test set.

(iii) I discarded a 26 km model-soup result, [23] because the combined checkpoints had already seen the full evaluation set, making the result contaminated.

9 Conclusion

The main lesson is that under a strict parameter budget, where the knowledge is stored matters more than architecture choice. Using parameters to learn a metric while keeping the map in the gallery worked better than trying to encode the map in the network. In hindsight, simple diagnostics should have been used earlier to identify this ceiling before further training. The remaining performance now depends more on gallery coverage than model capacity, suggesting that additional data is likely more valuable than a larger encoder.

References

[1] Sachin Mehta and Mohammad Rastegari. Mobilevit: Light-weight, general-purpose, and mobile-friendly vision transformer. In *International Conference on Learning Representations (ICLR)*, 2022.

[2] Tobias Weyand, Ilya Kostrikov, and James Philbin. Planet – photo geolocation with convolutional neural networks. In *European Conference on Computer Vision (ECCV)*, pages 37–55, 2016.

[3] Eric Müller-Budack, Kader Pustu-Iren, and Ralph Ewerth. Geolocation estimation of photos using a hierarchical model and scene classification. In *European Conference on Computer Vision (ECCV)*, pages 575–592, 2018.

[4] Paul Hongsuck Seo, Tobias Weyand, Jack Sim, and Bohyung Han. CPlaNet: Enhancing image geolocalization by combinatorial partitioning of maps. In *European Conference on Computer Vision (ECCV)*, pages 536–551, 2018.

[5] Lukas Haas, Michal Skreta, Silas Alberti, and Chelsea Finn. PIGEON: Predicting image geolocations. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12893–12902, 2024.

[6] Mingxing Tan and Quoc V. Le. Efficientnet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning (ICML)*, pages 6105–6114, 2019.

[7] Brandon Clark, Peter Kerrigan, Pranav P. Kulkarni, Viviana V. Cepeda, and Mubarak Shah. Where we are and what we're looking at: Query based worldwide image geo-localization using hierarchies and scenes. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 23182–23190, 2023.

[8] James Hays and Alexei A. Efros. IM2GPS: Estimating geographic information from a single image. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2008.

[9] Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. Convnext v2: Co-designing and scaling convnets with masked autoencoders. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16133–16142, 2023.

[10] Filip Radenović, Giorgos Tolias, and Ondrej Chum. Fine-tuning CNN image retrieval with no human annotation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41 (7):1655–1668, 2019.

[11] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

[12] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[13] Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. Randaugment: Practical automated data augmentation with a reduced search space. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[14] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[15] Leslie N. Smith and Nicholay Topin. Super-convergence: Very fast training of neural networks using large learning rates. In *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications*, volume 11006. SPIE, 2019.

[16] Hervé Jégou and Ondrej Chum. Negative evidences and co-occurences in image retrieval: The benefit of PCA and whitening. In *European Conference on Computer Vision (ECCV)*, pages 774–787, 2012.

[17] Alexis Conneau, Guillaume Lample, Marc'Aurelio Ranzato, Ludovic Denoyer, and Hervé Jégou. Word translation without parallel data. In *International Conference on Learning Representations (ICLR)*, 2018.

[18] Miloš Radovanović, Alexandros Nanopoulos, and Mirjana Ivanović. Hubs in space: Popular nearest neighbors in high-dimensional data. *Journal of Machine Learning Research*, 11: 2487–2531, 2010.

[19] Stephen Hausler, Sourav Garg, Ming Xu, Michael Milford, and Tobias Fischer. Patch-netvlad: Multi-scale fusion of locally-global descriptors for place recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14141–14152, 2021.

[20] Bingyi Cao, Andre Araujo, and Jack Sim. Unifying deep local and global features for image search. In *European Conference on Computer Vision (ECCV)*, pages 726–743, 2020.

[21] Ondrej Chum, James Philbin, Josef Sivic, Michael Isard, and Andrew Zisserman. Total recall: Automatic query expansion with a generative feature model for object retrieval. In *IEEE International Conference on Computer Vision (ICCV)*, 2007.

[22] Zhun Zhong, Liang Zheng, Donglin Cao, and Shaozi Li. Re-ranking person re-identification with k-reciprocal encoding. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1318–1327, 2017.

[23] Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, et al. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning (ICML)*, 2022.